

Date	July 1, 2026
Team ID	SB-AI-2026-9284
Project Name	Sustainable Waste Management AI Assistant
Maximum Marks	2 Marks

Technology Stack

Project Architecture Technologies

Below is the stack details chosen for the Sustainable Waste Management application:

- Frontend: React.js (Vite) - Component lifecycle management & fast compilation.
- Styling: Vanilla CSS3 - Dynamic custom HSL variables & theme styling.
- Backend: Flask (Python) - Lightweight REST router & simple CORS endpoints.
- Database: Google Firebase Firestore - NoSQL cloud logging & real-time updates.
- AI Core: Groq (Llama-3.3-70B) - High speed contextual inference with strict JSON schemas.
- Mapping: Leaflet.js & OSM - Offline-friendly maps with zero cost fees.
- Charts: Chart.js - HTML5 canvas rendering of statistics indicators.